

Term Paper Checkpoint 1: Systematic Literature Search

As it is still early in the term, you likely won't have settled on a topic for your term paper yet. This checkpoint is mostly constructed to help you start thinking about topics that may be interesting, and getting you comfortable with finding quality literature on a topic. We fully expect that many, if not most people's searches completed in this checkpoint will not make it to the final draft of their term paper, but the skills of searching databases and evaluating research will transfer well to later checkpoints.

Choosing a (Possible) Topic

Have a look at the list of term paper topics on Quercus. These are not by any means an exhaustive list of topics, they are mostly there to give you an idea of the *type* of topics you could write on. Most of the effort in this checkpoint will be you thinking about what topics are of interest to you, and what areas you might feel like researching this term. The eventual topic you pick will have a big impact on your course, so you'll want to choose carefully. But for this checkpoint, you can try out a topic that might be interesting, and be confident that you can easily switch before the next checkpoint.

Finding a Database

Through the library resources section on Quercus, you should be able to access a variety of databases on computing and other tangentially related topics. Think about your topic and which sorts of areas might be covered. These databases will lead you to research articles and published works, but they will not be the only sources of information you can use for your term paper. The point of searching these databases is to give you (and your reader) a thorough background of the research into the related topics. You will likely also need to use other sources like news articles and opinion pieces in your term paper, but that will come later.

Performing a Literature Search

Each database will have slightly different rules about syntax and querying. You will need to spend a bit of time reading up on each database's unique interface and possibly playing with them to see how they work. You should choose 2-3 databases on which to perform your search.

Once you're comfortable with your databases, your job is to write a query (or set of queries) that will get you the information you want with precision and recall.

- Precision: A measure of how much of the information returned is relevant to your topic.

- **Recall:** A measure of how much of the total relevant information that is available is returned by your query.

Precision without recall is useless (getting some relevant documents doesn't help if you don't know what else is out there), and recall without precision isn't much better (it's no good having every relevant document in the database if you're so overwhelmed with irrelevant documents that you can't find them)

Inclusion/Exclusion Criteria

You will need to think carefully about what criteria you want documents in your search to meet. Setting reasonable criteria can be a good way of improving recall without getting swamped by papers. You can set criteria on things like:

- Range of dates of publication
- Peer review status
- Opinion vs Evidence based publications
- Length/type of publication

In later stages, you'll be more discerning in your criteria, but at this stage, you want things that you can filter for within the database, or decide by quickly reading over the abstract of a paper.

Performing Your Search

Aim for a query + set of criteria that returns 10-20 documents that seem (at first glance) to be relevant across your databases. You may need to tweak either your query or criteria to get this to a reasonable level. If you're getting too many results, narrow your scope, if you're getting too few, add extra databases or loosen your restraints.

Again: the goal here isn't to get every possible piece of information you'll need to write your term paper, but to get a good representative set of publications that explore a topic or sub topic you may be interested in.

What to Submit

You will need to submit the following as a single pdf document:

- Your topic (explained in plain English)
- Your initial query (explained in plain English)
- A list of your chosen databases
- Your query, as translated into those databases (complete with date query was run, search parameters, and scope limitations)

- A list of paper titles and abstracts retrieved
- For any papers that were excluded based on their titles/abstracts, a brief 1 sentence explanation of why the paper was excluded
- A full bibliography for the retrieved papers (hint: most databases will allow you to download bib-tex versions of the paper references, or you can use a product like Zotero (<https://www.zotero.org>) to track and manage your bibliography... spend a bit of time learning these tools, you'll thank us by the end of the course)